

● AAROGYA GRID

The medicine is already there. Nobody can see it.

National medicine supply intelligence for India's primary health network — forecasting stock-outs across 2,816 facilities, and finding the stock sitting an hour away that would prevent them.

Build with AI: **Code for Communities**

Problem statement 03 — Smart Health & Supply Chain Resilience

Built with **Google Gemini**

THE PROBLEM

A sub-centre runs out of anti-snake venom in monsoon. The vials exist.

They are ninety minutes away, in a facility that will write them off unused in six weeks. Nobody knows either fact.

Both facilities report into a paper register that reaches the district office weeks later, if at all. India's public health supply chain does not primarily fail on procurement volume — it fails on **visibility**, and on **lateral movement**: the ability to see a shortage forming and move stock sideways before it becomes a stock-out.

4,594

POSITIONS AT ZERO STOCK

5,471

CRITICAL POSITIONS

1.92 M

UNITS EXPECTED SHORTFALL

₹11.4 L

STOCK HEADING TO EXPIRY

WHAT IT DOES

Four steps, from a spoken sentence to a signed dispatch order.

01

Capture

An ANM speaks, in Hindi, or photographs a paper register. Gemini turns it into structured stock data.

02

Forecast

Croston's method for intermittent demand, corrected for the censoring that stock-outs create in the ledger.

03

Rank risk

Monte Carlo over the resupply window returns stock-out probability and expected shortfall — not a point estimate.

04

Redistribute

Pairs facilities heading for expiry with facilities heading for zero. Names the batch. Prices the trip.

2,816

FACILITIES

128

DISTRICTS · 16 STATES

80,896

FACILITY × DRUG POSITIONS

232 M

PEOPLE IN CATCHMENT

A stock ledger records what was dispensed – not what was needed.

Once a facility hits zero, demand keeps arriving and stops being recorded. Fit naively on that ledger and you **systematically under-forecast exactly the facilities that are already failing** – then allocate them less, so they stock out again.

We fit with stocked-out periods excluded. Measured against the same estimator run on uncensored demand, so seasonality and recency cancel and only the censoring effect remains.

DISTRICT TIER	PAIRS	NAIVE	CORRECTED
Functioning	2,551	-2.7%	-0.8%
Strained	1,018	-4.7%	-0.8%
Disrupted	301	-10.0%	-2.6%

7.4 percentage points of forecast bias recovered in the worst-served districts — the ones where being wrong costs the most.

Not a dashboard alert. A dispatch order a storekeeper can act on.

GENERATED DISPATCH ORDER · BASTAR, CHHATTISGARH

DH Bastar-01 can spare 778 tablets (batch **B012-DH001**, expires in 850 days) while **SC Bastar-12** holds 0 tablets against a 100% chance of running short within its 15-day resupply window. Moving them **16 km** costs **₹743** and averts an expected **670.8 tablets** of unmet demand, cutting stock-out risk to 5%.

pick list: B012-DH001 × 781 (exp 2029-01-27) · risk reduction 95 pp

Every recommendation names the **specific batch and its expiry date**, and the quantity is bounded by what that batch physically holds. A recommendation a storekeeper cannot fill is not a recommendation.

Verbatim from the artefact the live site reads: `src/data/districts/DST-22-BASTAR.json`

Visibility is not the binding constraint. Unit economics is.

Of the needs the optimiser could not serve, we expected most to fail because the stock simply wasn't there. That is not what happens.

94.0% fail the benefit/cost gate. The medicine exists — only **1.5%** of declined needs fail for want of donor stock anywhere. It sits a median **35.6 km** away, and moving it costs more than the shortfall it averts because the model prices a separate vehicle trip per drug.

That reframes the intervention. Better dashboards do not fix this. **Pooled district-level logistics does** — one scheduled milk-run makes hundreds of individually uneconomic transfers economic at once.



2,798 accepted orders run over **2,083 distinct corridors**, and the planner never combines two orders onto one trip. Consolidate the corridor and the gate declining **13,672** needs is charging a truck per tablet.

Redistribution does not pay for itself in cash. It pays in lives.

NATIONAL PLAN	VALUE
Waste averted	+ ₹3.21 L
Transport cost	- ₹33.4 L
Net cash position	- ₹30.2 L
Shortfall averted	495,166 units

2,798 dispatches across 128 districts.

We could have shown a ₹2 crore headline. It would have been a shadow price on averted shortfall doing 115% of the work.

Instead the multiplier is **an explicit dial**. The plan breaks even when one unit of averted unmet demand is valued at ₹6.11. Whether a dose of a *Vital* medicine reaching a patient is worth that is a **ministry's judgement, not an engineer's** — so we surface the parameter rather than bury it in a headline.

The officer asks in Hindi. The model calls the tools. Every number is quoted, never computed.

LIVE • DISTRICT HEALTH OFFICER ASKS

"Bastar mein kaun se centre par dawa khatam hone wali hai?"

```
1 list_positions {limit:10} → 10 rows
2 district_status {} → 1 row
3 drug_reference {Ceftriaxone} → 1 row
4 drug_reference {Artesunate} → 1 row
answered from 4 calls • 7 facilities and 3 drugs cited • 22.2s
```

Nine tools, all pure functions over already-computed state. The **tool-call trace is shown in the interface** — a grounded answer nobody can check is indistinguishable from a confident guess.

NO IDS The model is never given a district code, facility id or drug id to emit. Asked about a district it had not been handed, it **invented the code "Lucknow"** — so arguments are natural-language names, resolved deterministically inside the tool.

NO MATHS It quotes numbers, never derives them. Testing caught it turning a 0.998 probability into "99.8%" — correct arithmetic, but **every ratio now ships a precomputed percent** to quote instead.

HAND LOOP The SDK's automatic function calling is deliberately unused: we reject unknown tool names, **Zod-validate every argument** before it touches data, and record the trace ourselves rather than asking the model what it did.

AUDITED Every numeric token in the model's prose is matched against the payloads it actually saw. The audit is **proven able to fail** by tampering a known-good answer. Result: **zero hallucinated numbers** across 8 scenarios.

Beds and people, on the same screen as the stock – because they explain it.

WORKFORCE	POSTS
Sanctioned	33,920
In position	26,981
Present today	22,232
Specialist vacancy	45.5%

Three levels, not one number. In rural India the **vacancy** gap and the **absence** gap are different problems with different fixes, and the distance between them is the finding.

That last number is the hinge. It does not rewrite any measured quantity — it **widens the error bar** around it. Three resources, one system.

BEDS	COUNT
Sanctioned	37,888
Functional	33,017
Staffable today	31,546
Occupancy · at capacity	83.3% · 100

219 stock-holding facilities have no pharmacist in post to keep the register. 393 facilities, covering 8.5 million people, carry stock figures nobody is in position to verify.

Four users, one dataset, four different questions.

ANM SUB-CENTRE	Speaks one sentence into a phone instead of filling a register by hand. Told immediately if what she reported looks wrong.
Storekeeper PHC / CHC	Receives a dispatch order naming a batch, a quantity, an expiry date and a distance. Nothing to interpret.
District Health Officer DISTRICT	Asks the grid a question in their own language and gets a grounded answer with its working shown. Sees what will fail in two weeks, and what can be moved to prevent it.
State & ministry NATIONAL	Sees which districts are structurally failing rather than unlucky, and where a logistics contract or a recruitment drive would buy more than more procurement.

DEPLOYABILITY & SCALE

One file stands between this and real data. The country scales linearly.

Everything downstream consumes one interface, FacilityDrugState. Swapping the simulator for a **DVDMS / e-Aushadhi** extract and the generated network for an **ABDM Health Facility Registry** pull changes **that one file and nothing else**.

THE INTEGRATION SEAM

```
sim/facilities.ts → ABDM HFR  
sim/inventory.ts → DVDMS extract  
sim/resources.ts → HRMIS / HMIS  
  
forecast/ · optimize/ · ai/ · app/ unchanged
```

SCOPE	DISTRICTS	NIGHTLY BATCH
This demo	128	95 s
One large state	~40	~30 s
All India, demo density	780	~10 min

Districts are evaluated independently with no shared state, so the work is **embarrassingly parallel** and trivially shardable by state — which is what makes federated, state-by-state deployment possible rather than aspirational.

Running on **Cloud Run, asia-south1**. Compute and inference are both in Mumbai and the container holds **no credential at all** — the service account attaches to the service. Nothing to rotate, leak, or revoke.

Stated plainly, because the honest answer is the stronger one.

REAL

Districts, their coordinates and state assignments. Facility tier structure, **bed norms and the sanctioned staffing establishment** per IPHS 2022. A 47-item catalogue — **46 drugs from India's National List of Essential Medicines** plus one tracked consumable — with VED classification, pack units, cold-chain flags and unit costs.

Real PHC inventory data is not public. Rather than claim otherwise, we built the seam that makes the swap a one-file change — and **every number in this deck is reproducible from a seed**, by anyone, from the public repository.

SIMULATED

The stock ledger, bed occupancy, staff attendance — **and district catchment populations**. All from a seeded, deterministic simulator parameterised from IPHS norms and published seasonality, **not fitted to observed data**. District and state rankings reflect a synthetic reliability parameter, not real performance.